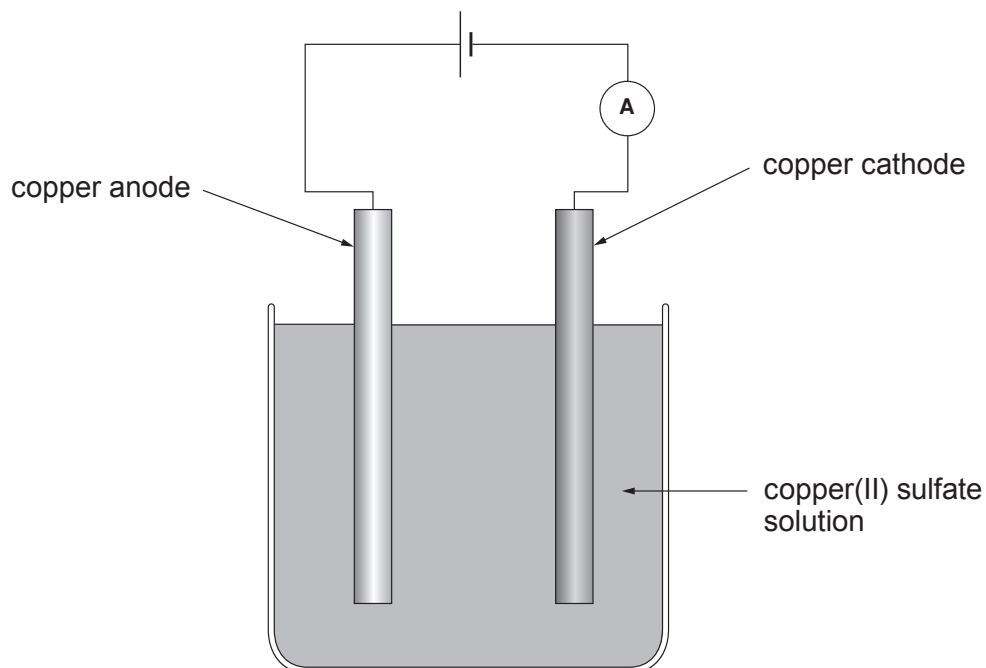


8. A student used the apparatus shown to investigate how changing current affects the mass of copper deposited on the cathode.



Equal volumes and concentrations of copper(II) sulfate solution were used each time. Each experiment was run for 10 minutes. All the readings were obtained at room temperature. The following procedure was followed before the mass of copper could be found.

- the electrode was carefully removed from the electrolyte
- the copper deposit was washed
- the electrode and copper deposit were dried

Her results are shown in the following table.

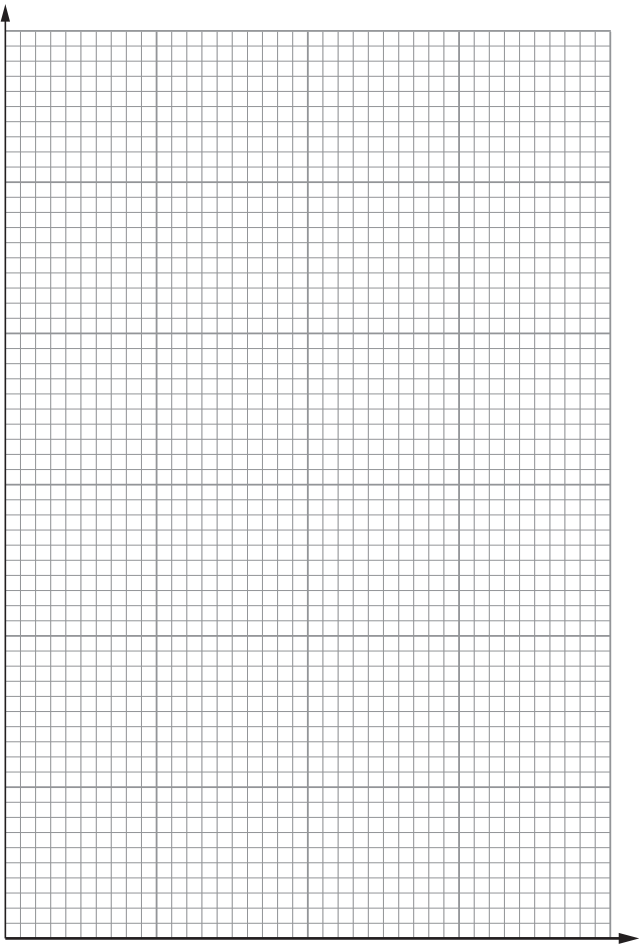
Current (A)	Mass of copper deposited (g)
0.0	0.00
0.5	0.16
1.0	0.30
1.5	0.43
2.0	0.60



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- (a) Choose suitable scales for the axes below and plot the current against the mass of copper deposited. Draw a suitable line. [4]

Mass of
copper
deposited
(g)



Current (A)

- (b) Use your graph to predict the mass of copper deposited using a current of 3.5 A. [1]

Mass of copper deposited = g



(c) Explain why the electrolyte keeps its blue colour during the electrolysis process. Include electrode equations to support your answer. [4]

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9

